

SPSS PROJECT REPORT

A Study on the Influence of Product Placement on Impulse Buying Behaviour Among College Students in Retail Stores

Statistical Analysis Using IBM SPSS

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1. PROJECT OVERVIEW

This project examines the influence of product placement on impulse buying behaviour among college students in retail stores. Primary data were collected through a structured questionnaire, and the responses were analyzed using IBM SPSS. The project focuses on relationships, predictive effects and mean comparisons using six statistical techniques.

Particular	Details
Research topic	A Study on the Influence of Product Placement on Impulse Buying Behaviour Among College Students in Retail Stores
Data source	Primary questionnaire responses
Number of responses	50
Questionnaire variables	30 study questions, in addition to basic respondent information
Software	IBM SPSS
Significance level	5% ($\alpha = 0.05$)
Main statistical techniques	Pearson correlation, Spearman rank correlation, Linear regression, Multiple regression, One-sample t-test, Paired-samples t-test

1.1 Objectives of the Study

- To examine the relationship between attractive product displays and impulse purchases.
- To examine the relationship between exceeding the planned shopping budget and regret from impulse purchases.
- To determine whether purchasing after prominent product placement predicts monthly impulse purchases.
- To examine the combined effect of selected product-placement variables on impulse purchases.
- To compare the overall product-placement influence score with the neutral test value.
- To compare the Product Placement Score and Impulse Buying Score.

1.2 Data Collection

The Excel response file contains 50 responses and 30 numbered study questions. Questions 1–20 are numerical/frequency or amount-based questions. Questions 21–30 are opinion statements recorded using a five-point Likert scale.

1.3 Likert Scale Coding

Response	Code
Strongly Disagree	1
Disagree	2
Neutral	3
Agree	4
Strongly Agree	5

The Likert responses were converted into numerical codes for SPSS analysis. The basic respondent information was collected for the questionnaire but was not used as an independent variable in the six selected inferential analyses.

2. ANALYSIS MAPPING

The following mapping follows the analysis plan shown in the submitted SPSS project. Each analysis has a clear identified purpose, variables, hypothesis, SPSS procedure and interpretation.

No.	Analysis	Variables	Purpose
1	Linear Correlation – Pearson	Q7 and Q19	To measure the relationship between attractive product-display attention and impulse purchases.
2	Rank Correlation – Spearman	Q13 and Q20	To measure the ranked relationship between exceeding budget and regret from impulse purchases.
3	Linear Regression	Q11 → Q19	To examine whether purchasing after prominent placement predicts impulse purchases.
4	Multiple Regression	Q7, Q11, Q17 → Q19	To examine the combined effect of selected product-placement variables on impulse purchases.
5	One-Sample t-Test	Q30, Test Value = 3	To compare the mean overall product-placement influence with the neutral value.
6	Paired-Samples t-Test	Product Placement Score ↔ Impulse Buying Score	To compare the two related composite scores.

3. GENERAL SPSS WORKFLOW

Prepare the questionnaire response file in Excel/CSV format.

Import the data into SPSS Data View.

Create variables in Variable View and assign suitable numeric measurement levels.

Convert the Likert responses in Q21–Q30 into numerical codes from 1 to 5.

Create the required composite scores for the paired-samples t-test.

Select the required analysis from the Analyze menu.

Move the mapped variables into the appropriate variable boxes.

Select the required statistical options, such as Pearson, Spearman, two-tailed testing, Enter method or Test Value = 3.

Run the analysis and examine the output tables.

Compare the p-value with 0.05 and write the interpretation.

Record the output in the report without changing the SPSS values.

4. LINEAR CORRELATION – PEARSON

A. Question / Research Problem

Find whether there is a significant relationship between attractive product displays catching attention and the number of impulse purchases per month.

B. Variables

Role	Variable
Independent / Pair 1	Q7 – How many times do attractive product displays catch your attention in a month?
Dependent / Pair 2	Q19 – How many impulse purchases do you make in a typical month?

C. Hypothesis

Hypothesis	Statement
H ₀ (Null)	There is no significant relationship between attractive product displays catching attention and impulse purchases per month.
H ₁ (Alternative)	There is a significant relationship between attractive product displays catching attention and impulse purchases per month.

D. SPSS Output – Key Values

Statistic	Value
Pearson Correlation (r)	–0.014
Sig. (2-tailed)	0.925
N	50

F. SPSS Procedure – Step by Step

- Open the SPSS data file.
- Create numeric variables X and Y and label them as Q7 and Q19.
- Enter/import the 50 responses into Data View.
- Go to Analyze → Correlate → Bivariate.
- Move Q7 and Q19 into the Variables box.
- Select Pearson and Two-tailed.
- Click OK to generate the output.

G. Interpretation

The Pearson correlation coefficient is -0.014 , indicating a very weak negative relationship between attractive product-display attention and impulse purchases. The significance value is 0.925 , which is greater than 0.05 . Therefore, the relationship is not statistically significant at the 5% level. The null hypothesis is not rejected.

H. Decision

Decision: Based on the reported Pearson correlation output, the p-value is greater than 0.05 . Therefore, the null hypothesis is not rejected.

Note: The numerical output in this section follows the submitted SPSS project output/report and is retained consistently throughout this document.

5. RANK CORRELATION – SPEARMAN

A. Question / Research Problem

Find the relationship between exceeding the planned shopping budget and regret from impulse purchases.

B. Variables

Role	Variable
Independent / Pair 1	Q13 – How many times do you exceed your planned shopping budget in a month?
Dependent / Pair 2	Q20 – How many times do you regret an impulse purchase in a month?

C. Hypothesis

Hypothesis	Statement
H ₀ (Null)	There is no significant relationship between exceeding the planned shopping budget and regret from impulse purchases.
H ₁ (Alternative)	There is a significant relationship between exceeding the planned shopping budget and regret from impulse purchases.

D. SPSS Output – Key Values

Statistic	Value
Spearman Correlation (ρ)	–0.101
Sig. (2-tailed)	0.486
N	50

F. SPSS Procedure – Step by Step

- Open the SPSS data file.
- Create variables X and Y and label them Q13 and Q20.
- Enter/import the data into Data View.
- Go to Analyze → Correlate → Bivariate.
- Move Q13 and Q20 into the Variables box.
- Select Spearman and Two-tailed.
- Click OK to obtain the correlation output.

G. Interpretation

The Spearman correlation coefficient is -0.101 , indicating a very weak negative relationship between exceeding the planned shopping budget and regretting impulse purchases. The significance value is 0.486 , which is greater than 0.05 . Therefore, the relationship is not statistically significant at the 5% level. The null hypothesis is not rejected.

H. Decision

Decision: Based on the reported spearman correlation output, the p-value is greater than 0.05 . Therefore, the null hypothesis is not rejected.

Note: The numerical output in this section follows the submitted SPSS project output/report and is retained consistently throughout this document.

6. LINEAR REGRESSION

A. Question / Research Problem

Find whether purchasing a product after noticing its prominent placement significantly influences the number of impulse purchases made per month.

B. Variables

Role	Variable
Independent / Pair 1	Q11 – How many times do you purchase a product after noticing its prominent placement?
Dependent / Pair 2	Q19 – How many impulse purchases do you make in a typical month?

C. Hypothesis

Hypothesis	Statement
H_0 (Null)	Purchasing after prominent placement does not significantly influence impulse purchases per month.
H_1 (Alternative)	Purchasing after prominent placement significantly influences impulse purchases per month.

D. SPSS Output – Key Values

Statistic	Value
R	0.268
R Square	0.072
Adjusted R Square	0.052
Std. Error of the Estimate	0.759
F	3.699
Sig.	0.060
Constant B	1.808
Q11 B	0.236
Q11 Beta	0.268
Q11 t	1.923
Q11 Sig.	0.060

E. Statistical Equation

$$Q19 = 1.808 + 0.236(Q11)$$

F. SPSS Procedure – Step by Step

- Open the SPSS data file.
- Create numeric variables X and Y and label them Q11 and Q19.
- Enter/import the responses into Data View.
- Go to Analyze → Regression → Linear.
- Move Q19 into the Dependent box.
- Move Q11 into the Independent(s) box.
- Keep Method as Enter.
- Click OK to generate Model Summary, ANOVA and Coefficients tables.

G. Interpretation

The regression model has $R = 0.268$ and $R^2 = 0.072$. This indicates that approximately 7.2% of the variation in impulse purchases is explained by purchasing after noticing prominent placement. The ANOVA significance value is 0.060, which is greater than 0.05. Therefore, the regression relationship is not statistically significant at the 5% level. The null hypothesis is not rejected.

H. Decision

Decision: Based on the reported regression output, the p-value is greater than 0.05. Therefore, the null hypothesis is not rejected.

Note: The numerical output in this section follows the submitted SPSS project output/report and is retained consistently throughout this document.

7. MULTIPLE REGRESSION

A. Question / Research Problem

Find attractive product displays, purchase after prominent placement and promotional displays examining products significantly influence impulse purchases per month.

B. Variables

Role	Variable
Independent / Pair 1	Independent variables: Q7, Q11 and Q17
Dependent / Pair 2	Q19 – How many impulse purchases do you make in a typical month?

C. Hypothesis

Hypothesis	Statement
H ₀ (Null)	Attractive display attention, purchase after prominent placement and examining products due to placement do not significantly influence impulse purchases per month.
H ₁ (Alternative)	Attractive display attention, purchase after prominent placement and examining products due to placement significantly influence impulse purchases per month.

D. SPSS Output – Key Values

Statistic	Value
R	0.281
R Square	0.079
Adjusted R Square	0.019
Std. Error of the Estimate	0.772
Durbin-Watson	2.178
F	1.313
Sig.	0.282
Constant B	1.968
Q7 B	–0.003
Q7 Beta	–0.008
Q7 Sig.	0.956
Q11 B	0.262
Q11 Beta	0.296
Q11 Sig.	0.053
Q17 B	–0.089
Q17 Beta	–0.090
Q17 Sig.	0.549

E. Statistical Equation

$$Q19 = 1.968 - 0.003(Q7) + 0.262(Q11) - 0.089(Q17)$$

F. SPSS Procedure – Step by Step

- Open the SPSS data file.
- Create numeric variables for Q7, Q11, Q17 and Q19.
- Enter/import the data into Data View.
- Go to Analyze → Regression → Linear.
- Move Q19 into the Dependent box.
- Move Q7, Q11 and Q17 into the Independent(s) box.
- Keep Method as Enter.
- Select the required statistics such as ANOVA, Coefficients and Durbin-Watson.
- Click OK for the output.

G. Interpretation

The multiple regression model has $R = 0.281$ and $R^2 = 0.079$, indicating that approximately 7.9% of the variation in impulse purchases is explained jointly by Q7, Q11 and Q17. The overall model significance is 0.282, which is greater than 0.05. Therefore, the overall regression model is not statistically significant at the 5% level. The null hypothesis is not rejected.

H. Decision

Decision: Based on the reported regression output, the p-value is greater than 0.05. Therefore, the null hypothesis is not rejected.

Note: The numerical output in this section follows the submitted SPSS project output/report and is retained consistently throughout this document.

8. ONE-SAMPLE T-TEST

A. Question / Research Problem

Find whether the overall influence of product placement on impulse buying behaviour differs significantly from the neutral value.

B. Variables

Role	Variable
Independent / Pair 1	Q30 – Overall, product placement influences my impulse buying behaviour.
Dependent / Pair 2	Test Value = 3 (Neutral)

C. Hypothesis

Hypothesis	Statement
H_0 (Null)	There is no significant difference between the sample mean and the test value 3.
H_1 (Alternative)	There is a significant difference between the sample mean and the test value 3.

D. SPSS Output – Key Values

Statistic	Value
N	50
Mean	3.06
Std. Deviation	0.956
Std. Error Mean	0.135
T	0.444
Df	49
Sig. (2-tailed)	0.659
Mean Difference	0.060

F. SPSS Procedure – Step by Step

- Open the SPSS data file.
- Create the Q30 variable and code the five-point Likert responses from 1 to 5.
- Enter/import the responses into Data View.
- Go to Analyze → Compare Means → One-Sample T Test.
- Move Q30 into the Test Variable(s) box.
- Enter 3 as the Test Value because 3 represents Neutral.
- Click Options/Statistics if required and retain the default confidence level.
- Click OK to generate the output.

G. Interpretation

The sample means 3.06 and the test value is 3. The t-value is 0.444 with 49 degrees of freedom, and the significance value is 0.659. Since 0.659 is greater than 0.05, there is no statistically significant difference between the sample mean and the neutral value of 3. The null hypothesis is not rejected.

H. Decision

Decision: Based on the reported T-test output, the p-value is greater than 0.05. Therefore, the null hypothesis is not rejected.

Note: The numerical output in this section follows the submitted SPSS project output/report and is retained consistently throughout this document.

9. PAIRED-SAMPLES T-TEST

A. Question / Research Problem

Find whether there is a significant difference between the Product Placement Score and Impulse Buying Score.

B. Variables

Role	Variable
Independent / Pair 1	Pair 1 – Product Placement Score
Dependent / Pair 2	Pair 2 – Impulse Buying Score

C. Hypothesis

Hypothesis	Statement
H ₀ (Null)	There is no significant difference between the Product Placement Score and Impulse Buying Score.
H ₁ (Alternative)	There is a significant difference between the Product Placement Score and Impulse Buying Score.

D. SPSS Output – Key Values

Statistic	Value
Product Placement Score Mean	2.94
Product Placement Score SD	0.45
Impulse Buying Score Mean	2.92
Impulse Buying Score SD	0.50
Paired correlation	0.473
Correlation Sig.	0.001
Mean Difference	0.02
SD of Difference	0.49
T	0.29
Df	49
Sig. (2-tailed)	0.776

F. SPSS Procedure – Step by Step

- Open the SPSS data file.
- Create numeric variables for the items used in the Product Placement Score and Impulse Buying Score.
- Use Transform → Compute Variable to calculate the composite Product Placement Score.
- Use Transform → Compute Variable to calculate the composite Impulse Buying Score.
- Go to Analyze → Compare Means → Paired-Samples T Test.
- Select Product Placement Score as Variable 1 and Impulse Buying Score as Variable 2.
- Move the pair into the Paired Variables box.
- Click OK to obtain Paired Samples Statistics, Paired Samples Correlations and Paired Samples Test.

G. Interpretation

The mean Product Placement Score is 2.94, while the mean Impulse Buying Score is 2.92. The mean difference is 0.02. The paired samples t-test gives $t = 0.29$ with 49 degrees of freedom and a significance value of 0.776. Since 0.776 is greater than 0.05, the difference between the two related scores is not statistically significant. The null hypothesis is not rejected. The paired correlation of 0.473 with $p = 0.001$ is a separate correlation result and should not be confused with the p-value of the paired t-test.

H. Decision

Decision: Based on the reported T-test output, the p-value is greater than 0.05. Therefore, the null hypothesis is not rejected.

Note: The numerical output in this section follows the submitted SPSS project output/report and is retained consistently throughout this document.

10. SUMMARY OF STATISTICAL FINDINGS

Analysis	Variables	Main Result	p-value	Decision
Pearson correlation	Q7 & Q19	$r = -0.014$	0.925	H ₀ not rejected
Spearman correlation	Q13 & Q20	$\rho = -0.101$	0.486	H ₀ not rejected
Linear regression	Q11 → Q19	$R^2 = 0.072$	0.060	H ₀ not rejected
Multiple regression	Q7, Q11, Q17 → Q19	$R^2 = 0.079$	0.282	H ₀ not rejected
One sample T-test	Q30 vs 3	$t = 0.444$	0.659	H ₀ not rejected
Paired T-test	Placement Score vs Impulse Score	$t = 0.29$	0.776	H ₀ not rejected

11. OVERALL FINDINGS

- The Pearson correlation indicates a very weak negative relationship between attractive product-display attention and impulse purchases, and the relationship is not statistically significant.
- The Spearman analysis indicates a very weak negative relationship between exceeding the planned shopping budget and regret from impulse purchases, and the relationship is not statistically significant.
- The linear regression model explains 7.2% of the variation in impulse purchases, but the overall regression is not statistically significant at the 5% level.
- The multiple regression model explains 7.9% of the variation in impulse purchases, but the combined model is not statistically significant at the 5% level.
- The one-sample t-test shows that the mean response for the overall product-placement influence statement does not differ significantly from the neutral value of 3.
- The paired-samples t-test shows that the Product Placement Score and Impulse Buying Score do not differ significantly.

12. CONCLUSION

The study used six statistical techniques to examine different aspects of product placement and impulse buying behaviour. Based on the reported SPSS outputs, none of the six tested relationships or mean differences is statistically significant at the 5% significance level. The results therefore provide no statistical evidence in this sample to reject the respective null hypotheses.

13. LIMITATIONS OF THE STUDY

- The analysis is based on 50 responses.
- The study is limited to the respondents included in the questionnaire data.
- The findings are based on self-reported responses.
- Only the variables selected in the analysis mapping were used for inferential testing.
- The study uses a 5% significance level for hypothesis testing.

14. SUGGESTED REPORT FORMAT FOR SUBMISSION

Cover page

Project overview

Objectives of the study

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Analysis mapping

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Spearman rank correlation

Linear regression

Multiple regression

One sample T-test

Paired-samples T-test

Summary of findings

Overall conclusion

Limitations

Appendix – questionnaire / raw-data reference / SPSS output screenshots

15. APPENDIX: VARIABLE MAPPING

Variable	Description
Q7	Attractive product displays catch my attention / catch attention in a month
Q11	Purchase after noticing prominent placement
Q13	Exceed planned shopping budget in a month
Q17	Promotional displays make me examine a product before buying it
Q19	Impulse purchases in a typical month
Q20	Regret an impulse purchase in a month
Q30	Overall, product placement influences my impulse buying behaviour